

Arduino-Based Object Detection System Using Ultrasonic Sensor and LCD Display

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Abstract

This paper presents the design and implementation of an object detection system using an Arduino Uno, ultrasonic sensor, buzzer, and LCD display. The system is capable of detecting objects within a specified distance threshold (10 cm) and alerting the user through an audible buzzer and a visual LCD message[2]. The ultrasonic sensor measures the distance between the sensor and an object, while the Arduino processes this data and triggers corresponding outputs[1],[3]. When an object is detected within 10 cm, the system activates the buzzer and displays "Object Detected" on the LCD along with the measured distance. When no object is present within this range, the LCD displays "No Object" with the current distance. The system is low-cost, simple to implement, and suitable for basic automation and safety applications.

Keywords

Arduino Uno, Ultrasonic Sensor, Object Detection, Distance Measurement, LCD Display, Buzzer Alarm, Embedded System.

INTRODUCTION

In recent years, automation and sensing technologies have played a significant role in improving safety and efficiency across various domains. Object detection is one such crucial application that finds use in robotics, obstacle avoidance systems, security, and industrial

automation[2]. The ultrasonic sensor has become a popular choice for distance measurement due to its simplicity, low cost, and non-contact nature[4]. This project demonstrates a compact and efficient object detection system built around the Arduino Uno microcontroller. The system continuously monitors the distance between the sensor and any obstacle using ultrasonic pulses[2],[4]. Based on the measured distance, the Arduino provides real-time feedback via an LCD display and an audible buzzer alarm.

Objective

The object detector using Arduino is a simple project that uses sensors to detect objects in front of it.

The main objectives of this project are:

- ◆ Object detection: To sense the presence of an object within a specified range using sensors.
- ◆ Automatic indication/Alert : To give alerts when an object is detected in a defined range.
- ◆ Hands on learning of Arduino : To understand interfacing of Arduino with input sensors(Ultrasonic,IR,PIR) and output devices(LCD,LED,Buzzer motor).

◆ Practical Applications : To develop a working prototype that can be applied in:

- Obstacle detection for robots.
- Parking assistance system.
- Security systems.
- Object counter.

with appropriate distance/range to run this project through which the ultrasonic sensor by receiving the instructions to be followed while working. The instructions to the connections of TriggerPin, echo and buzzer were also inputted in this project.

❖ Pruthviraj .S. Gadhe (24BEE0125):

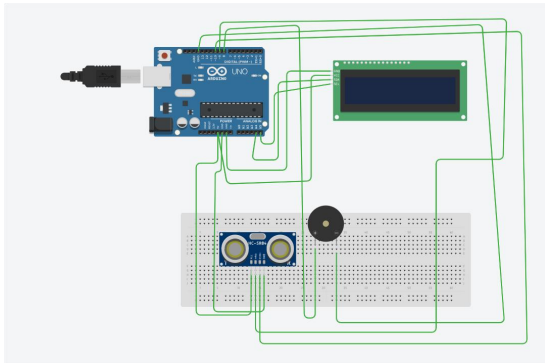
Contribution

❖ Gautam .A. Harshe (24BEE0141):

Worked with the Arduino coding, covering the software part of this project. Code was written

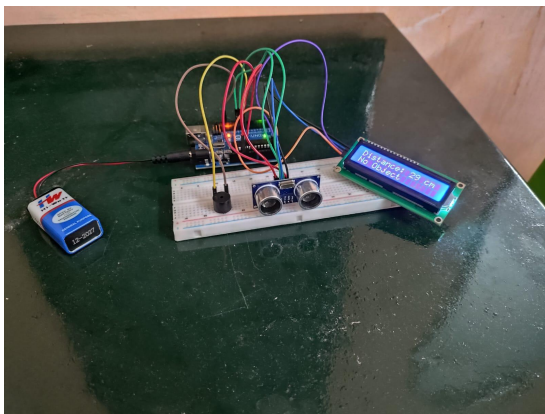
The hardware connections were made by connecting wires across the appropriate ends of the GND, echo and TriggerPins through the Arduino UNO R3, buzzer and ultrasonic sensor. Also the required connections are done across the digital and analog pins on the Arduino.

Block Design



METHODOLOGY

The system consists of an Arduino Uno, HC-SR04 Ultrasonic Sensor, 16x2 LCD Display, Buzzer, Breadboard, Battery, and Connecting Wires.

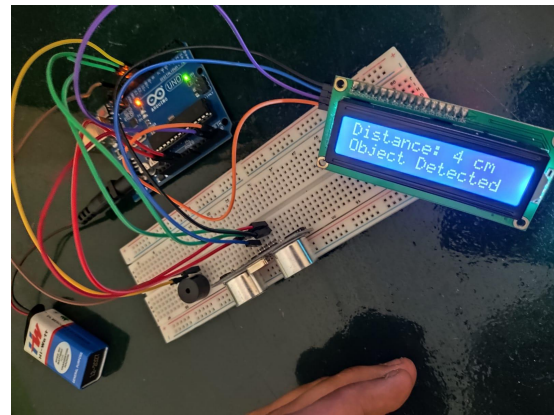


(ii) In the above figure the object is out of range and the buzzer is not ringing. The LCD is displaying the distance and "No Object".

A. Working Principle

1. The ultrasonic sensor emits high-frequency sound waves.
2. These waves reflect back when they hit an object.
3. The sensor measures the time taken for the echo to return and

- sends this data to the Arduino[3],[4].
4. The Arduino calculates the distance using the formula: $\text{Distance (cm)} = (\text{Time} \times 0.034) / 2$ [1],[3].
 5. If the distance is less than or equal to 10 cm, the Arduino triggers the buzzer and displays "Object Detected" with the distance on the LCD.
 6. If the distance is greater than 10 cm, the buzzer remains off, and the LCD displays "No Object" with the distance.



(i) The above figure shows the buzzer ringing when the object is within 10 cm and LCD displaying the distance of object from the ultrasonic sensor and also "Object Detected".

B. Hardware Connections

- HC-SR04 Ultrasonic Sensor: VCC → 5V, GND → GND, TRIG → D9, ECHO → D10
- Buzzer: D8
- LCD (I2C): SDA → A4, SCL → A5
- Power Supply: 9V Battery

C. Software Implementation

The system was programmed using Arduino IDE. The logic continuously reads sensor data, calculates the distance, and updates the LCD display. The buzzer is controlled using conditional statements based on the measured distance[4].

RESULTS AND INFERENCE

The prototype was successfully tested. The system accurately measured distances and responded correctly to the defined threshold of 10 cm.

Test Condition	Measured Distance (cm)	LCD Message	Buzzer Status
Object at 5 cm	5	Object Detected	ON
Object at 10 cm	10	Object Detected	ON
Object at 20 cm	20	No Object	OFF
No Object	>50	No Object	OFF

The results indicate high reliability in object detection within the 2–400 cm sensor range. The system performed consistently under indoor lighting and temperature conditions[1],[3].

FUTURE SCOPE

The present system can be enhanced in several ways:

- Integration with IoT for remote monitoring and alert systems.
- Use of multiple ultrasonic sensors for 360° obstacle detection.
- Implementation in robotic vehicles for automatic obstacle avoidance.
- Addition of servo motors to

automatically track detected objects.

- Integration with camera modules for visual verification and AI-based detection.

Goals

SDG 9 – Industry, Innovation and Infrastructure

The project promotes **innovation** through electronics and automation.

Using Arduino, sensors, and buzzers encourages **technological learning** and the creation of **smart, automated systems**.

Such systems can be expanded for **smart vehicles, robotics, and industrial safety**.

The project represents affordable innovation that can enhance industrial processes safely.

SDG 11 – Sustainable Cities and Communities

Object detection systems can be used for **smart parking, automatic doors, and traffic management** in smart cities.

They help make urban spaces **safer and more efficient**.

Example: Parking sensors reduce collisions and make transportation safer in crowded cities.

SDG 3- Good Health and Well-being

Modified versions of this project can be used for **assistive devices** for visually impaired people (e.g., obstacle detection with audio alerts).

Can prevent **accidents** by alerting people or machines when an object is too close.

This contributes to the goal of ensuring healthy lives and promoting well-being.

SDG 4 - Quality Education

This project helps students **learn practical STEM (Science, Technology, Engineering, and Mathematics) skills**.

Encourages hands-on learning and creativity in electronics and programming.

It supports education that prepares learners for technological careers.

SDG 12 - Responsible Consumption and Production

By using low-cost, energy-efficient components like Arduino and sensors, it encourages **sustainable use of materials and electronics**.

Can also help in **waste management systems** (e.g., smart bins detecting object levels).

CONCLUSION

The Arduino-based object detection system offers a simple, low-cost, and effective method for detecting objects within a defined range. It combines ultrasonic sensing with a real-time display and audible alert, making it suitable for educational, industrial, and automation applications. The project successfully demonstrates how microcontroller-based systems can be used for reliable proximity detection and alert mechanisms.

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